

**CSA0612-DESIGN AND ANALYSIS OF  
ALGORITHMS.**

**CO1-ASSESSMENT TOOL 1-APPLICATION  
BASED QUESTIONS.**

**Q75: MULTIMEDIA STREAMING PLATFORM.**

A multimedia streaming platform divides large media data into two equal parts and processes them recursively, performing extra linear work at each stage. The recurrence relation is:

$$T(n) = 2T(n/2) + n$$

This represents a divide-and-conquer approach where the problem of size  $n$  is split into two subproblems of size  $n/2$ , and additional linear work  $n$  is done to merge or process results. This is similar to how streaming services handle encoding, buffering, or parallel data processing.

**Solution Using Master Theorem:**

The recurrence relation fits the form:

$$T(n) = aT(n/b) + f(n)$$

Here:

- $a = 2$ (two subproblems)
- $b = 2$ (problem size halved each time)
- $f(n) = n$ (extra linear work per level)

Now compute:  $\log_b a = \log_2 2 = 1$ .

Compare  $f(n)$  with  $n^{\log_b a}$ :

- $f(n) = n$
- $n^{\log_b a} = n^1 = n$

Since  $f(n) = \Theta(n^{\log_b a})$ , by Case 2 of the Master Theorem:

$$T(n) = \Theta(n \log n)$$

### **Time Complexity:**

$$T(n) = O(n \log n)$$

This means the streaming platform's recursive processing grows proportionally with  $n \log n$ , which is efficient for large-scale data handling.

### **Space Complexity:**

Recursion depth  $= \log_2 n$ . At each level, temporary storage may require  $O(n)$ . Thus, overall Space Complexity  $= O(n)$ .

**Final Answer:**

The multimedia streaming platform recurrence relation  $T(n) = 2T(n/2) + n$  resolves to:

- Time Complexity:  $O(n \log n)$
- Space Complexity:  $O(n)$

This shows the system is efficient, scalable, and well-suited for real-world streaming applications.

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## **Q76: INSURANCE CLAIM VERIFICATION SYSTEM**

An insurance claim verification system must efficiently process and validate claims submitted by policyholders. The system divides the verification process into multiple sub-tasks such as checking policy details, validating claim documents, and cross-verifying with external records. Each sub-task contributes to the overall complexity of the system. The recurrence relation is given as:

$$T(n) = T(n - 1) + O(1)$$

This indicates that each claim verification depends on the previous claim, with constant additional work at each step.

### **Solution:**

The recurrence relation can be solved as follows:

$$T(n) = T(n - 1) + O(1)$$

Expanding step by step:

$$\begin{aligned} T(n) &= T(n - 1) + 1 \\ &= T(n - 2) + 1 + 1 \\ &= T(n - 3) + 1 + 1 + 1 \end{aligned}$$

$$= \dots = T(1) + (n - 1)$$

Thus,

$$T(n) = O(n)$$

### **Time Complexity:**

The time complexity of the insurance claim verification system is  $O(n)$ . This means the system scales linearly with the number of claims, making it predictable and manageable for large datasets.

### **Space Complexity:**

Since the recurrence relation only depends on the previous claim and requires constant additional work, the space usage is minimal. Therefore, Space Complexity =  $O(1)$ .

### **Final Answer:**

The insurance claim verification system recurrence relation  $T(n) = T(n - 1) + O(1)$  resolves to:

- Time Complexity:  $O(n)$
- Space Complexity:  $O(1)$

This demonstrates that the system is efficient, requiring linear time to process claims while using constant space, making it suitable for large-scale insurance operations.

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